

Indian Knowledge System

Number System and Unit of Measurement

UNIT 2

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Number System in India- Historical Evidence

- The contribution of ancient Indians to the development of mathematical concepts is well known and acknowledged.
- Several archaeological excavations provide evidence about maturity and supremacy of Indian mathematical foundations.
- An inscription on a temple wall in Gwalior dating back to the ninth century CE is considered the oldest recorded example of a zero.

Number System in India- Historical Evidence

- Still, surviving Sanskrit texts reveal a rich tradition of Indian mathematical discoveries lasting more than 2,500 years.
- In the Early Vedic period (1200–600 bc), a decimal system of numbers was already established in India, together with rules for arithmetical operations (ganita) and geometry (rekha-ganita).

Salient Features of the Indian Numeral System

- The Indian numeral system has a long history; the origin and the evolution of the numbers could be traced from the time of the Vedic period.
- Being an oral tradition, unique and unambiguous names were to be attributed to the numbers.
- The Sanskrit language has unique names for numbers starting from one and going up to very large numbers; the first nine digits have unique names (ekam, dve, trini, catvari, panca, sat, sapta, asta, nava).

Salient Features of the Indian Numeral System

Name	Indian notation	Power notation	Indian system	Short scale Western
एक (<i>ēka</i>)	1	10^0	One	One
दश (<i>daśa</i>)	10	10^1	Ten	Ten
शत (<i>śata</i>)	100	10^2	One hundred	One hundred
सहस्र (<i>sahasra</i>)	1,000	10^3	One thousand	One thousand
लक्ष (<i>lakṣa</i>)	1,00,000	10^5	One lakh	One hundred thousand
कोटि (<i>kōṭi</i>)	1,00,000 <i>śata</i>	10^7	One crore	Ten million
शङ्कु (<i>śaṅku</i>)	1,00,000 <i>koṭi</i>	10^{12}	Ten kharab or One lakh crore	One trillion
महाशङ्कु (<i>mahāśaṅku</i>)	1,00,000 <i>śaṅku</i>	10^{17}	One shankh or One thousand crore crore	One hundred quadrillion
वृन्द (<i>vṛnda</i>)	1,00,000 <i>mahāśaṅku</i>	10^{22}		Ten sextillion (ten trilliard)
महावृन्द (<i>mahāvṛnda</i>)	1,00,000 <i>vṛnda</i>	10^{27}		One octillion
पद्म (<i>padma</i>)	1,00,000 <i>mahāvṛnda</i>	10^{32}		One hundred nonillion
महापद्म (<i>mahāpadma</i>)	1,00,000 <i>padma</i>	10^{37}		Ten undecillion
खर्व (<i>kharva</i>)	1,00,000 <i>mahāpadma</i>	10^{42}		One tredecillion
महाखर्व (<i>mahākharva</i>)	1,00,000 <i>kharva</i>	10^{47}		One hundred quattuordecillion
समुद्र (<i>samudra</i>)	1,00,000 <i>mahākharva</i>	10^{52}		Ten sexdecillion
ओघ (<i>ogha</i>)	1,00,000 <i>samudra</i>	10^{57}		One octodecillion
महौघ (<i>mahaugha</i>)	1,00,000 <i>ogha</i>	10^{62}		One hundred novemdecillion

The Concept of Zero and Its Importance

- The concept of zero was established during the period 500-300 BCE.
- Ancient Indians used a decimal System that allowed them to develop a method for handling large numbers.
- Brahmagupta developed a symbol for zero in 628 CE.
- With this invention zero could be used as an independent numeral for computational purposes.

Large Numbers and their representation

- Ancient Indians developed a systematic approach to the number- names that they used to describe large numbers.
- Three categories of naming conventions were employed to develop the number names in Sanskrit:
 - i. All numbers in the digits from 0 to 9 had unique names (sunya, ekam, dve, trini, catvari, panca, sat, sapta, asta, nava).
 - ii. All numbers in the range of 11 to 99 had an additive principle for naming.
 - iii. All numbers of higher powers of 10 starting from 10^2 were named using multiplicative principle using the unit digits as a factor for multiplication.

Unique approaches to Represent Numbers

- Ancient Indian mathematics seamlessly integrated mathematics with literature and poetry.
- We find well developed systems to address this requirement and discuss two of them here:
 - i. Bhuta-Samkhya system is a system of expressing numbers by means of words representing certain entities.
 - ii. Katapayadi system employs a technique to convert the numerals to alphabets using certain rules.

Bhūtasamkhyā (भूतसंख्या) system

- Bhūtasamkhyā (भूतसंख्या) system is a method of recording numbers in Sanskrit using common nouns having connotations of numerical values.
- The method was introduced already in astronomical texts in antiquity, but it was expanded and developed during the medieval period.
- A kind of rebus system, *bhūtasamkhyā* has also been called the “concrete number notation”.

Bhūtasamkhyā (भूतसंख्या) system

TABLE 6.3 Word Numerals for Bhūta-samkhyā System

Number	Represented by (partial list only)*
0	śūnya, anata, pūrṇa, kha
1	ādi, candra, prithivī, eka
2	aśvin, pairs of limbs, ayana, dvandva, dvi
3	rāma, guṇa, loka, kāla, agni, trinetra
4	veda, śruti, yuga, aśrama, varṇa, samudra, kṛta
5	bhūta, śāstra, bāṇa, pāṇḍava, indriya
6	aṅga, ṛtu, darśana, ṣaṇmukha, ṣaṭ
7	ṛṣi, adri, svara, dhātu, chandas
8	vasu, bhujaṅga, siddhi, dik, kuñjara, nāga
9	gṛha, aṅka, nanda
10	dik, angulī, avatāra, rāvaṇaśiras
11	rudra
12	āditya, rāśi
13	viśva, kāma
14	manu
15	tithi, dina

* For all the items listed, any word from the synonyms may be used

Katapayadi System

- Another System to convert the numerals to words is to associate a number to one or more alphabets.
- Using the alphabets in place of numbers, one can construct words, which by deciphering one alphabet at a time will reveal the number expressed in the word.
- The advantage of such a system lies in representing large numbers using a word, which can be easily remembered.
- This provided a very efficient method of presenting results of complex calculations using number symbols.

Katapayadi System

The assignment of letters to the numerals are as per the following arrangement.

1	2	3	4	5	6	7	8	9	0
ka ക ക	kha ഖ ഖ	ga ग ഗ	gha घ ഘ	nga ङ ങ	ca च ച	cha छ ഛ	ja ज ജ	jha झ ഝ	nya ञ ഞ
ṭa ट ട	ṭha ठ ഠ	ḍa ड ഡ	ḍha ढ ഢ	ṇa ण ണ	ta त ത	tha थ ഥ	da द ദ	dha ध ധ	na न ന
pa प പ	pha फ ഫ	ba ब ബ	bha भ ഭ	ma म മ	-	-	-	-	-
ya य യ	ra र ര	la ल ല	va व വ	śha श ശ	sha ष ഷ	sa स സ	ha ह ഹ	-	-

Measurement For Time, Distance and Weight

- In the ancient Indian literature three fundamental physical measures for quantifying length, time and weight are found.
- There are several sources in the literature specifying units of measurement for these three physical quantities.
- Three generic measures pertaining to length, weight and time could be used to measure physical entities.

Notion of Paramanu

- At the smallest level, there is a fundamental measure called paramanu.
- The measure of Paramanu are as follows:

Paramanu (length): 2.88×10^{-7} mm

Paramanu (weight): 5.79×10^{-5} g

Paramanu (time): 1.31×10^{-5} seconds

Measures for Length

TABLE 6.5 Ancient Indian Measures for Length

Unit	Multiplier of Preceding Unit*	No. of Paramāṇus	Length (in mm)
Paramāṇu-rajas	1	1	2.8778×10^{-7}
Rena	7	7	2.0145×10^{-6}
Truti	7	49	1.4101×10^{-5}
Vāthyaṇa-rajas	7	343	9.8709×10^{-5}
Śaśa-rajas	7	2,401	6.9096×10^{-4}
Edaka-rajas	7	16,807	4.8367×10^{-3}
Go-rajas	7	1,17,649	0.033857202
Likṣā-rajas	7	8,23,543	0.237000411
Sarapa	7	57,64,801	1.65900288
Yava	7	4,03,53,607	11.61302016
Anguli-parva	7	28,24,75,249	81.29114114

Adapted from Ifrah, G. (2004). *The Universal History of Numbers II*, Penguin Books, pp. 138-139.

Unit	Multiplier of Preceding Unit*	No. of Angulas	Length (in metre)
Angula	1	1	0.016764
Dhanurmuṣṭi	8	8	0.134112
Prājāpatya-hasta	3	24	0.402336
Dhanus	4	96	1.609344
Gārhapatya-dhanus	1.125	108	1.810512
Goruta	2000	216,000	3621.024
Yojana	4	864,000	14484.096

* The length of an Angula is 16.764 mm as per Indus inch.

THANKS